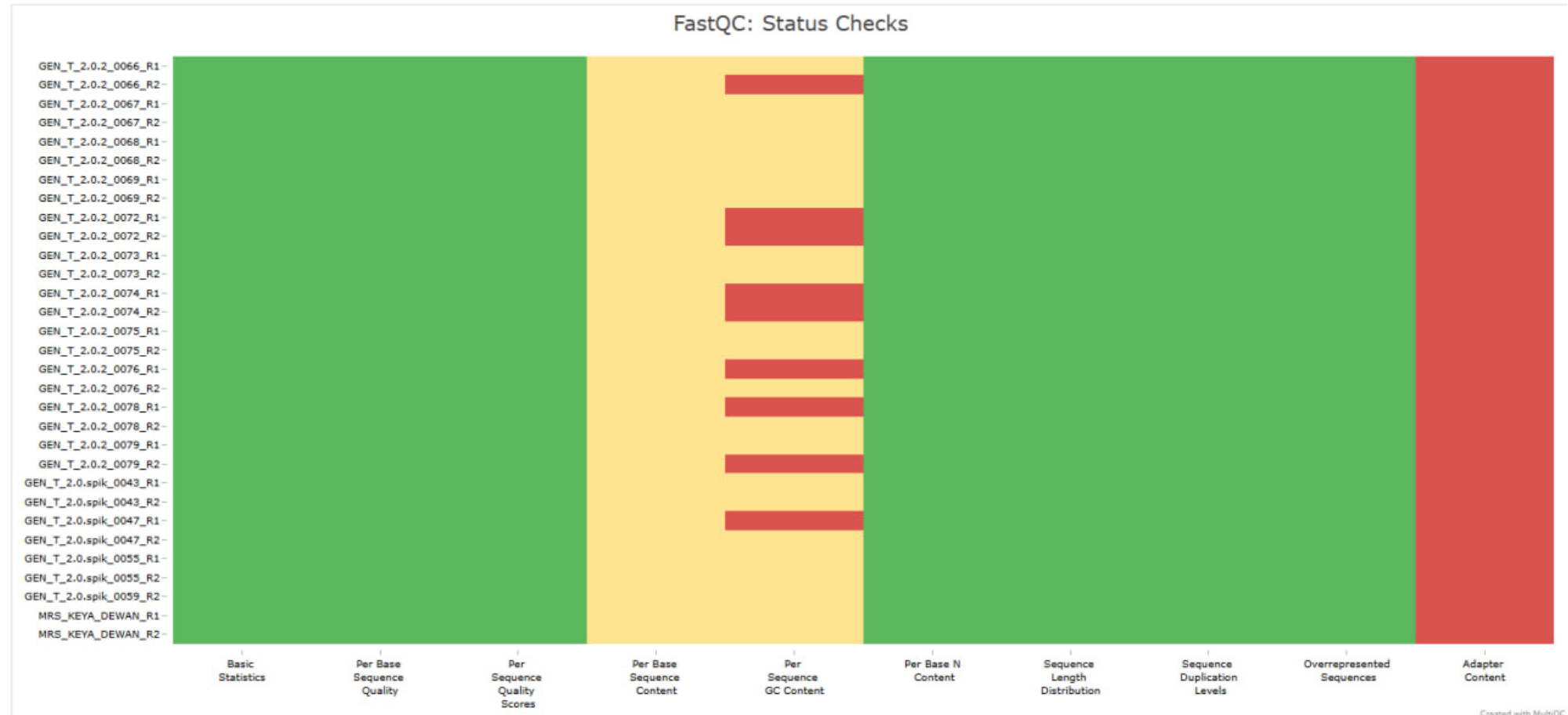


Run 10 Result Consolidation

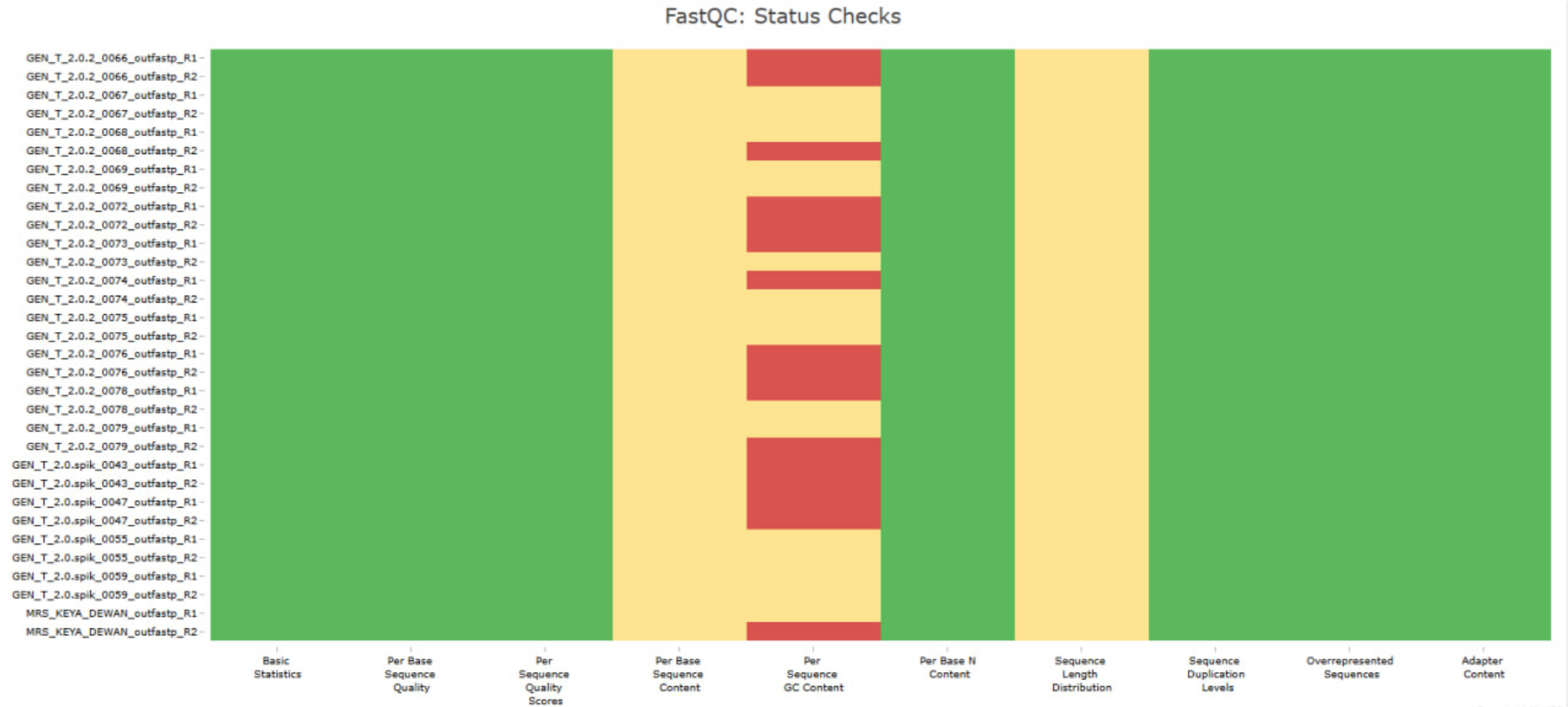
Gender Table

Sample_ID	Sample_Name	Gender In TRF	Gender in IGV
TP-09-LIB01	BABY. MRITHYOUNJOY SIL (3Y M)	Male	Male
TP-09-LIB02	MRS. MAHALAKSHMI S (30Y F)	Female	Female
TP-09-LIB03	MASTER. SAI ISHAAN (11Y M)	Male	Male
TP-09-LIB04	MR. MURUGAN [1619] (43Y M)	Male	Male
TP-12-LIB05	MRS. NAGA SUDHA C (30Y F) - POC	Male (confirmed from MCC)	Male
TP-12-LIB06	MRS. GOWDICHERLA. LAKSHMI MANASA - (HEG 36668) (29Y F)	Female	Female
TP-12-LIB07	MR. GOWDICHERLA HARI CHARAN SHARMA (38Y M)	Male	Male
TP-12-LIB08	MR. SUBRAMANIYAN S (47Y M)	Male	Male
TP-12-LIB09	MR. RAMESH KUMAR (62Y M)	Male	Male
TP-12-LIB10	MASTER. ARNAV GOHIL (13Y M)	Male	Male
TP-12-LIB11	MR. DHEERAJ KUMAR (36Y M)	Male	Male
TP-12-LIB12	MRS. RICHA (35Y F)	Female	Female
TP-13-LIB13	MASTER. NAVIN AMUTHAN B (8Y M)	Male	Male
TP-13-LIB14	MRS. KOTHA REVATHI - (HEGC005145) (34Y F)	Female	Female
TP-13-LIB15	MR. DHANAMJAYA RAO - (HEG C005146) (40Y M)	Male	Male
TP-13-LIB16	MRS. KEYA DEWAN (48Y F) (no TRF)	Female	Female

Before Fastp Multiqc



After Fastp Multiqc

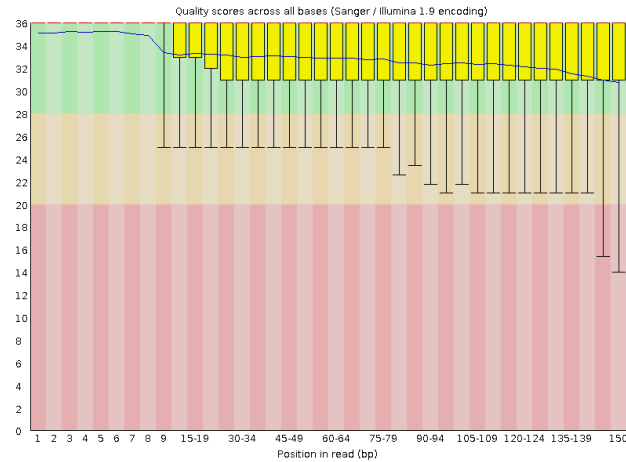


TP-09-LIB01 Before and After Fastqc

Before:

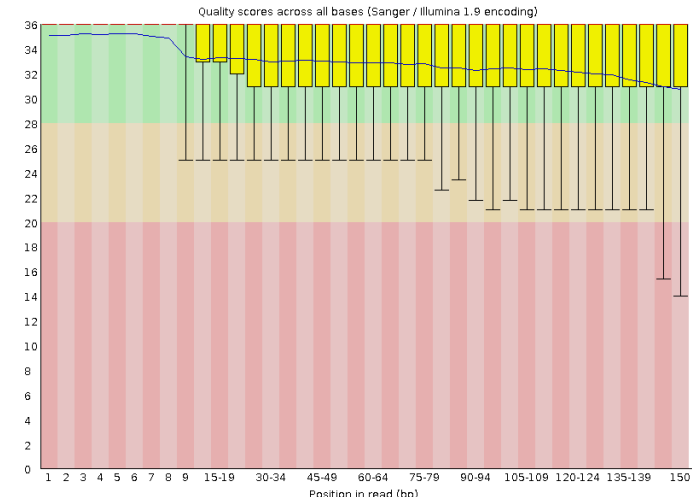
R1:

✔ Per base sequence quality



R2:

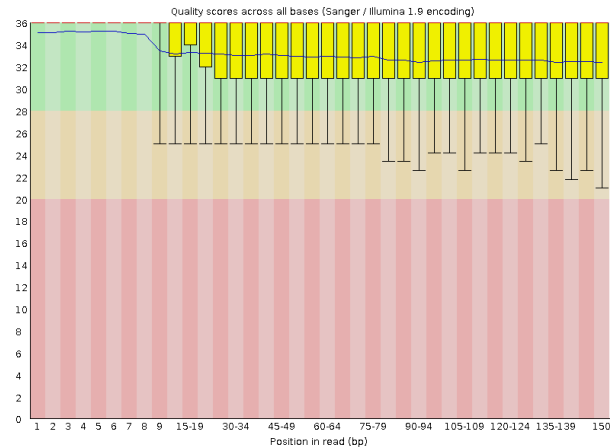
✔ Per base sequence quality



After:

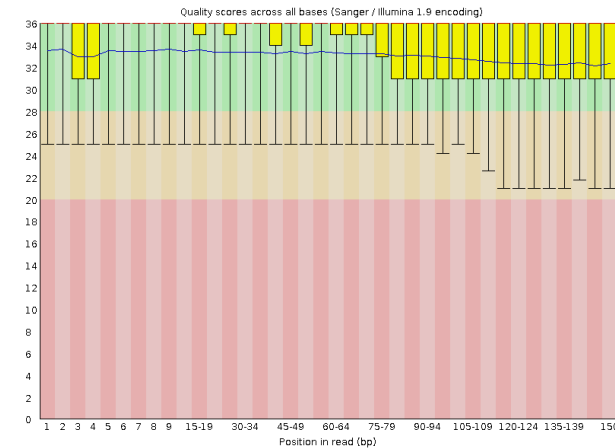
R1:

✔ Per base sequence quality



R2:

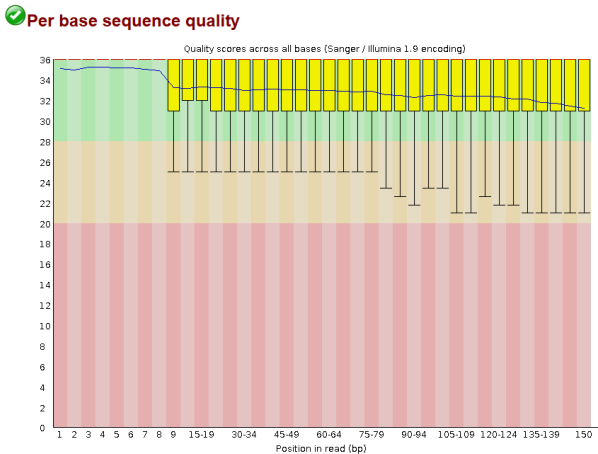
✔ Per base sequence quality



TP-09-LIB02Before and After Fastqc

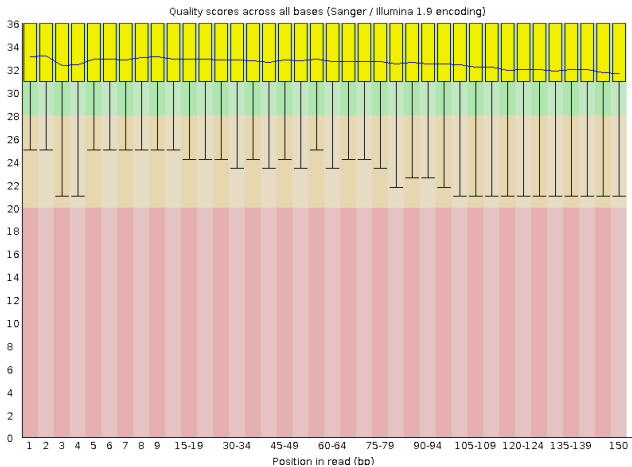
Before:

R1:



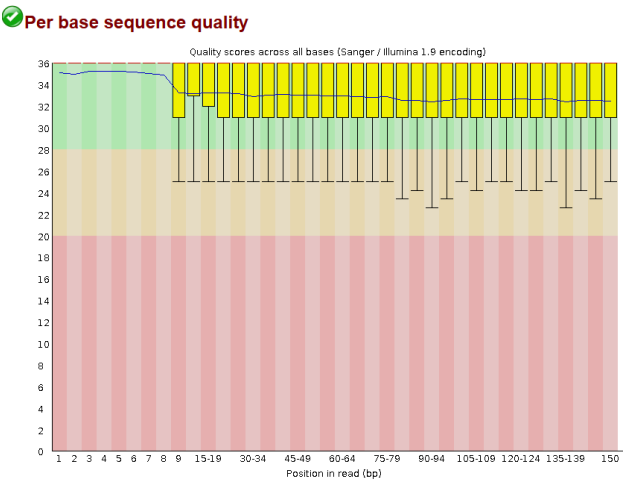
R2:

✔ Per base sequence quality



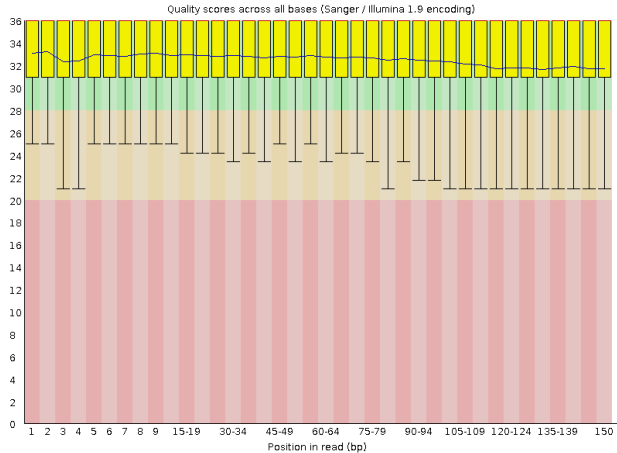
After:

R1:



R2:

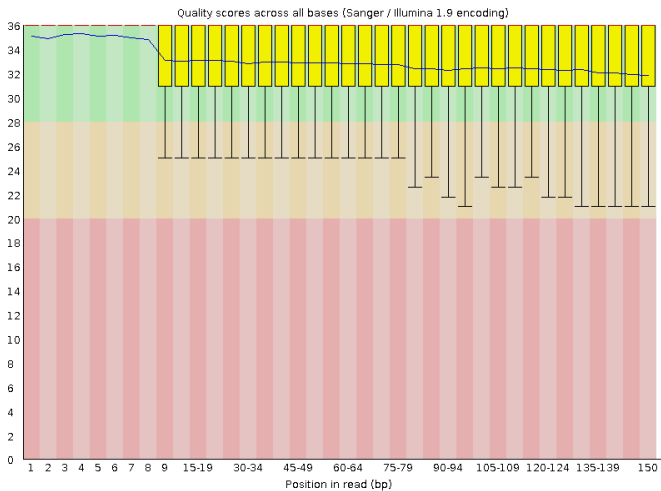
✔ Per base sequence quality



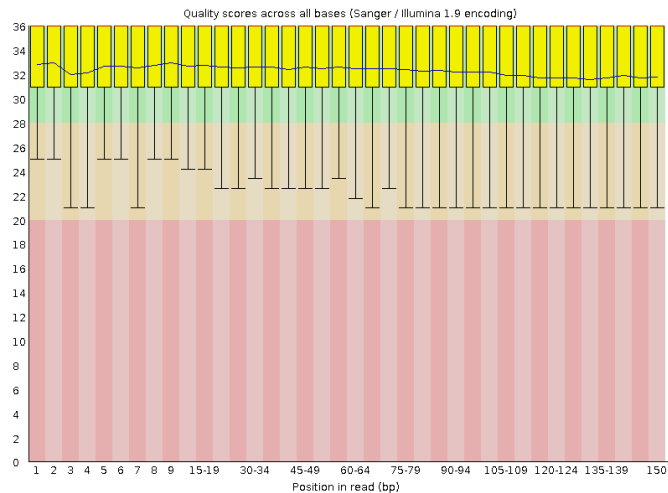
TP-13-LIB15 Before and After Fastqc

Before:

R1:  Per base sequence quality

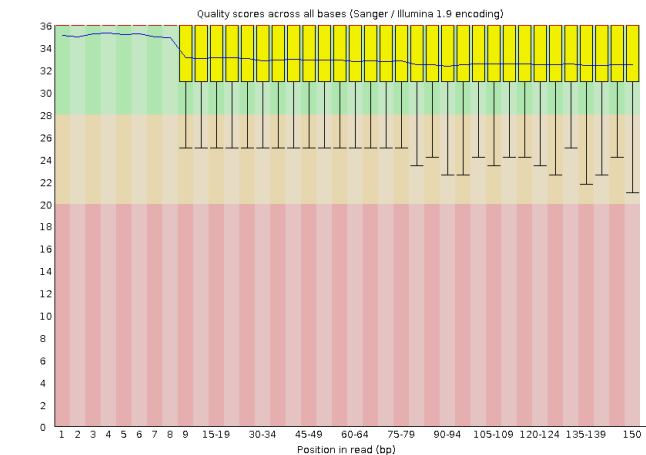


R2:  Per base sequence quality



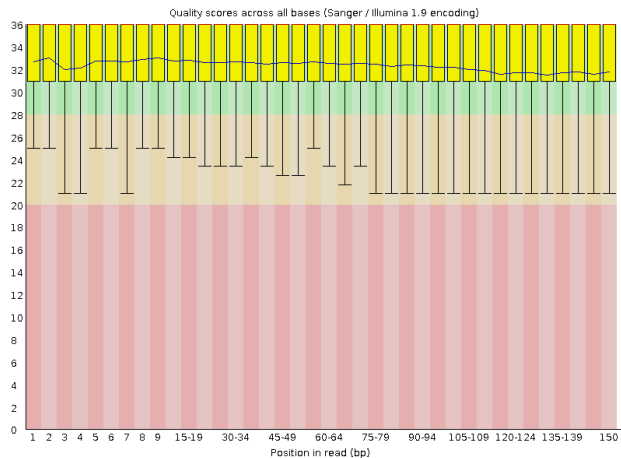
After:

R1:  Per base sequence quality



R2:

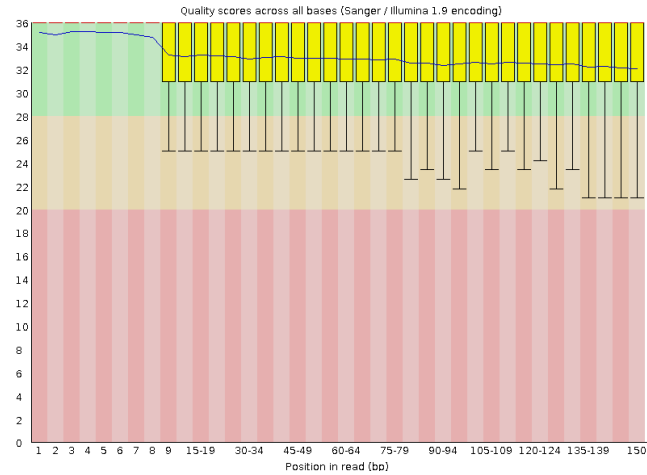
 Per base sequence quality



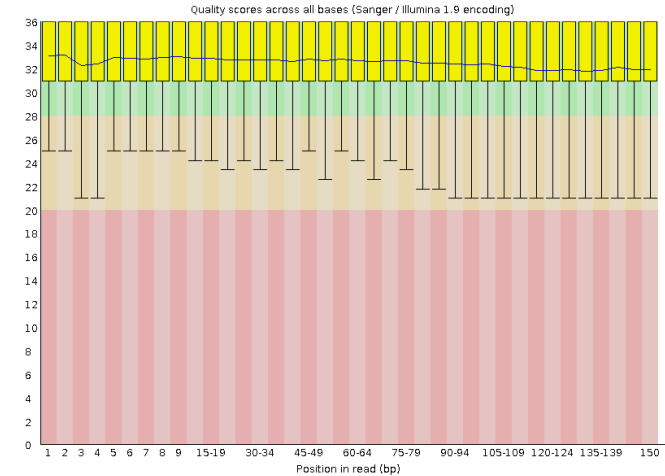
TP-13-LIB14 Before and After Fastqc

Before:

R1:  **Per base sequence quality**

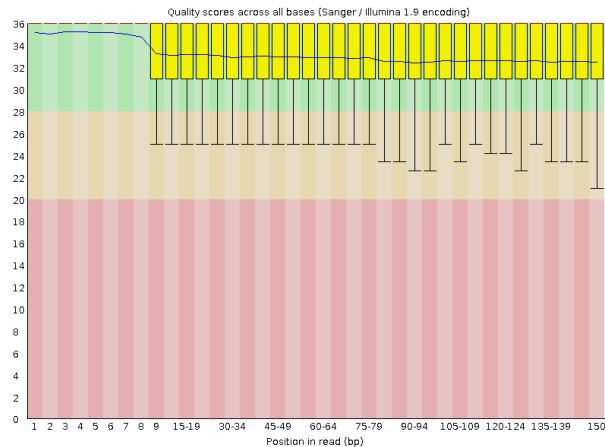


R2:  **Per base sequence quality**

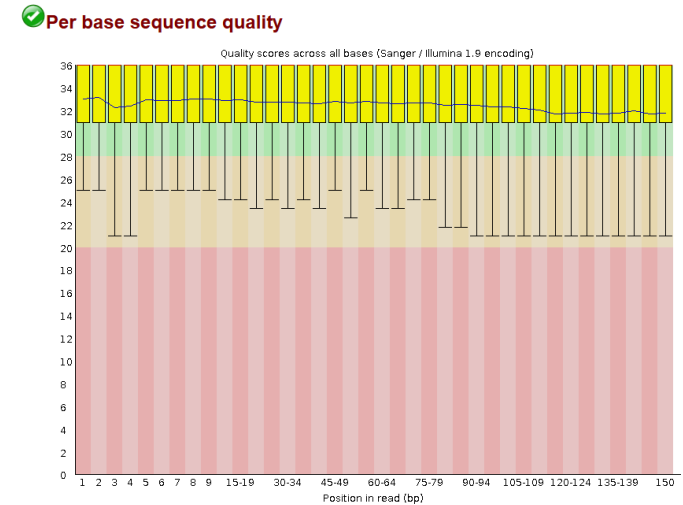


After:

R1:  **Per base sequence quality**



R2:



Fastp Result

FASTP COMPARISSION RUN 8										
BEFOREFASTP										
Sample_ID	Sample_Name		NUMBER OF READS	SIZE OF READS	Q30	TOTAL DATA GENERATED	NUMBER OF READS	SIZE OF READS	Q30	TOTAL DATA GENERATED
TP-09-LIB01	BABY. MRITHYOUNJOY SIL (3Y M)	GEN_T_2.0.spik_0043	49.748116 M	150bp	86.218055%	7.462217 G	49.569490 M	130bp	86.875089%	6.490902 G
TP-09-LIB02	MRS. MAHALAKSHMI S (30Y F)	GEN_T_2.0.spik_0047	52.922914 M	150bp	85.171193%	7.938437 G	52.578218 M	135bp	85.718202%	7.134984 G
TP-09-LIB03	MASTER. SAI ISHAAN (11Y M)	GEN_T_2.0.spik_0055	71.494822 M	150bp	85.648362%	10.724223 G	71.100376 M	135bp	86.217466%	9.609447 G
TP-09-LIB04	MR. MURUGAN [1619] (43Y M)	GEN_T_2.0.spik_0059	48.778100 M	150bp	85.751191%	7.316715 G	48.604446 M	131bp	86.374103%	6.394478 G
TP-12-LIB05	MRS. NAGA SUDHA C (30Y F) - POC	GEN_T_2.0.2_0066	69.606486 M	150bp	84.578325%	10.440973 G	69.085620 M	138bp	85.065246%	9.588071 G
TP-12-LIB06	MRS. GOWDICHERLA. LAKSHMI MANASA - (HEG 36668) (29Y F)	GEN_T_2.0.2_0067	66.911074 M	150bp	85.467361%	10.036661 G	66.438728 M	141bp	85.818446%	9.391791 G
TP-12-LIB07	MR. GOWDICHERLA HARI CHARAN SHARMA (38Y M)	GEN_T_2.0.2_0068	94.547190 M	150bp	85.645620%	14.182078 G	93.936990 M	139bp	86.041677%	13.110084 G
TP-12-LIB08	MR. SUBRAMANIYAN S (47Y M)	GEN_T_2.0.2_0069	67.712538 M	150bp	86.029545%	10.156881 G	67.304212 M	141bp	86.421985%	9.537720 G
TP-12-LIB09	MR. RAMESH KUMAR (62Y M)	GEN_T_2.0.2_0072	90.432122 M	150bp	86.059064%	13.564818 G	89.875546 M	140bp	86.425915%	12.668827 G
TP-12-LIB10	MASTER. ARNAV GOHIL (13Y M)	GEN_T_2.0.2_0073	80.044182 M	150bp	85.341065%	12.006627 G	79.510170 M	139bp	85.818828%	11.060476 G
TP-12-LIB11	MR. DHEERAJ KUMAR (36Y M)	GEN_T_2.0.2_0074	69.935478 M	150bp	85.355083%	10.490322 G	69.509502 M	141bp	85.706100%	9.801832 G
TP-12-LIB12	MRS. RICHA (35Y F)	GEN_T_2.0.2_0075	71.027402 M	150bp	83.955181%	10.654110 G	70.348620 M	141bp	84.355751%	9.955823 G
TP-13-LIB13	MASTER. NAVIN AMUTHAN B (8Y M)	GEN_T_2.0.2_0076	70.222572 M	150bp	84.094572%	10.533386 G	69.727692 M	141bp	84.454132%	9.859302 G
TP-13-LIB14	MRS. KOTHA REVATHI - (HEGC005145) (34Y F)	GEN_T_2.0.2_0078	63.079748 M	150bp	85.117793%	9.461962 G	62.610480 M	142bp	85.457877%	8.926295 G
TP-13-LIB15	MR. DHANAMJAYA RAO - (HEG C005146) (40Y M)	GEN_T_2.0.2_0079	74.664404 M	150bp	84.668939%	11.199661 G	74.106130 M	143bp	85.032823%	10.598228 G
TP-13-LIB16	MRS. KEYA DEWAN (48Y F)	MRS. KEYA DEWAN	70.083270 M	150bp	85.136858%	10.512491 G	69.628662 M	142bp	85.515284%	9.925530 G

HS Metrics Twist Spikein Bed

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000				
BAIT_SET	BAIT			
	TWIST SPIK RUN = 10			
	TP-09-LIB01	TP-09-LIB02	TP-09-LIB03	TP-09-LIB04
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	37453133	37453133	37453133	37453133
TARGET_TERRITORY	37453133	37453133	37453133	37453133
BAIT_DESIGN_EFFICIENCY	1	1	1	1
TOTAL_READS	49569490	52578218	71100376	48604446
PF_READS	49569490	52578218	71100376	48604446
PF_UNIQUE_READS	49569490	52578218	71100376	48604446
PCT_PF_READS	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1
PF_UQ_READS_ALIGNED	49565554	52573983	71094885	48600799
PCT_PF_UQ_READS_ALIGNED	0.999921	0.999919	0.999923	0.999925
PF_BASES_ALIGNED	6459694752	7114642000	9567434912	6367053738
PF_UQ_BASES_ALIGNED	6459694752	7114642000	9567434912	6367053738
ON_BAIT_BASES	3687187189	3892737360	5285073705	3661954556
NEAR_BAIT_BASES	2000434333	2347182205	3144804733	1940305410
OFF_BAIT_BASES	772073230	874722435	1137556474	764793772
ON_TARGET_BASES	2712970494	2787163942	3783209425	2692876091
PCT_SELECTED_BASES	0.880478	0.877053	0.881101	0.879883
PCT_OFF_BAIT	0.119522	0.122947	0.118899	0.120117
ON_BAIT_VS_SELECTED	0.648283	0.623844	0.626945	0.653657
MEAN_BAIT_COVERAGE	98.448	103.936	141.112	97.7743
MEAN_TARGET_COVERAGE	72.4364	74.4174	101.012	71.8999
PCT_USABLE_BASES_ON_BAIT	0.568055	0.545585	0.549987	0.572675
PCT_USABLE_BASES_ON_TARGET	0.417965	0.390634	0.393697	0.421125
FOLD_ENRICHMENT	49.0335	47.0015	47.4532	49.4065
ZERO_CVG_TARGETS_PCT	0.0127519	0.0131027	0.0119321	0.0133424
PCT_EXC_DUPE	0	0	0	0
PCT_EXC_MAPQ	0.0562626	0.0520243	0.0529737	0.0560085
PCT_EXC_BASEQ	0.0188189	0.0197639	0.0190083	0.0198942
PCT_EXC_OVERLAP	0.39184	0.359565	0.365679	0.405535
PCT_EXC_OFF_TARGET	0.748274	0.739112	0.736278	0.747912
FOLD_80_BASE_PENALTY	1.72468	1.51872	1.68353	1.63409
PCT_TARGET_BASES_0X	1.7869	1.8364	1.7052	1.8652
PCT_TARGET_BASES_1X	0.982131	0.981636	0.982948	0.981348
PCT_TARGET_BASES_2X	0.980816	0.98032	0.981735	0.980057
PCT_TARGET_BASES_10X	0.974625	0.97531	0.977396	0.974241
PCT_TARGET_BASES_20X	0.956578	0.966126	0.970741	0.958708
PCT_TARGET_BASES_30X	0.906808	0.939808	0.955635	0.916041
PCT_TARGET_BASES_40X	0.817451	0.878395	0.923876	0.834231
PCT_TARGET_BASES_50X	0.703348	0.778112	0.869881	0.722297
PCT_TARGET_BASES_100X	0.220884	0.207562	0.46641	0.20376
HS_LIBRARY_SIZE	0	0	0	0
HS_PENALTY_10X	0	0	0	0
HS_PENALTY_20X	0	0	0	0
HS_PENALTY_30X	0	0	0	0
HS_PENALTY_40X	0	0	0	0
HS_PENALTY_50X	0	0	0	0
HS_PENALTY_100X	0	0	0	0
AT_DROPOUT	0.0845982	0.0898103	0.0827615	0.0888206
GC_DROPOUT	13.5417	9.95415	13.0645	11.5827

HS Metrics Twist 2.0.2

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000													
BAIT													
BAIT_SET	T_2.0.2_RUN = 10												
	TP-12-LIB05	TP-12-LIB06	TP-12-LIB07	TP-12-LIB08	TP-12-LIB09	TP-12-LIB10	TP-12-LIB11	TP-12-LIB12	TP-13-LIB13	TP-13-LIB14	TP-13-LIB15	TP-13-LIB16	
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	
TOTAL_READS	69085620	66438728	93936990	67304212	89875546	79501070	69509502	70348620	69727692	62610480	74106130	69628662	
PF_READS	69085620	66438728	93936990	67304212	89875546	79501070	69509502	70348620	69727692	62610480	74106130	69628662	
PF_UNIQUE_READS	69085620	66438728	93936990	67304212	89875546	79501070	69509502	70348620	69727692	62610480	74106130	69628662	
PCT_PF_READS	1	1	1	1	1	1	1	1	1	1	1	1	
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1	1	1	1	1	
PF_UQ_READS_ALIGNED	69077947	66434617	93931074	67301507	89870454	79505441	69505349	70342691	69722680	62606826	74101344	69624591	
PCT_PF_UQ_READS_ALIGNED	0.999889	0.999938	0.999937	0.999996	0.999943	0.999941	0.999996	0.999916	0.999928	0.999942	0.999935	0.999942	
PF_BASES_ALIGNED	9578734883	9388840644	13098235035	9536169330	12660331515	11047353534	9792037791	9943227540	9846584819	8922487875	10590745996	9916417330	
PF_UQ_BASES_ALIGNED	9578734883	9388840644	13098235035	9536169330	12660331515	11047353534	9792037791	9943227540	9846584819	8922487875	10590745996	9916417330	
ON_BAIT_BASES	5874536315	5587121290	7997073234	5711265630	7723065255	6790617682	5956749046	5921398068	5312973264	6291959317	5977472122	5977472122	
NEAR_BAIT_BASES	2374740141	2497585782	3307108401	2484414558	3261879138	2745490911	2519455141	2623680984	2477164876	2323183431	2774453573	2568841631	
OFF_BAIT_BASES	1329458427	1304133572	1794053400	1340489142	1675387122	1511244941	1315833604	1337233710	1448021875	1286331180	1524333106	1370103577	
ON_TARGET_BASES	3650852946	3418989133	4916289066	3491450307	4723064965	4205656338	3628706622	3603003280	3558420764	3200758438	3756632955	3590711710	
PCT_SELECTED_BASES	0.861207	0.861097	0.863031	0.859431	0.867666	0.865622	0.865622	0.855833	0.852942	0.855833	0.856069	0.861835	
PCT_OFF_BAIT	0.138793	0.138903	0.136969	0.140569	0.132334	0.136797	0.134378	0.134487	0.147058	0.144167	0.143931	0.138165	
ON_BAIT_VS_SELECTED	0.712127	0.691073	0.707444	0.696863	0.703059	0.712095	0.702761	0.695133	0.705049	0.695765	0.693986	0.699421	
MEAN_BAIT_COVERAGE	135.828	129.183	184.904	132.053	178.569	157.009	137.729	138.32	136.912	122.844	145.479	138.208	
MEAN_TARGET_COVERAGE	100.138	93.7782	134.847	95.7657	129.547	115.355	99.5304	98.8254	97.6026	87.7924	103.039	98.4883	
PCT_USABLE_BASES_ON_BAIT	0.612692	0.594894	0.609994	0.598908	0.609612	0.613953	0.607718	0.600886	0.60059	0.595205	0.59368	0.602232	
PCT_USABLE_BASES_ON_TARGET	0.38077	0.38404	0.375001	0.369063	0.37281	0.380242	0.370207	0.361899	0.369092	0.358576	0.354459	0.361765	
FOLD_ENRICHMENT	45.6225	44.268	45.4184	44.5925	45.3793	45.7263	44.7564	44.7355	44.7564	44.2961	44.195	44.8411	
ZERO_CVG_TARGETS_PCT	0.00802981	0.00786076	0.00711061	0.00713526	0.0071423	0.00713878	0.00726909	0.0077868	0.00726557	0.00795937	0.00691691	0.00817773	
PCT_EXC_DUPE	0	0	0	0	0	0	0	0	0	0	0	0	
PCT_EXC_MAPQ	0.0555805	0.0495884	0.0520495	0.0519619	0.0515118	0.0523683	0.0514877	0.0512715	0.0533684	0.0506947	0.0506217	0.0505565	
PCT_EXC_BASEQ	0.0203393	0.0184087	0.0185516	0.0171506	0.0178857	0.0190355	0.018989	0.0207995	0.0206092	0.0188188	0.0193668	0.0189	
PCT_EXC_OVERLAP	0.354078	0.330103	0.353949	0.340368	0.349271	0.360572	0.348942	0.337667	0.35084	0.336825	0.336018	0.342519	
PCT_EXC_OFF_TARGET	0.737436	0.737365	0.734012	0.731497	0.730809	0.737976	0.730141	0.729332	0.724617	0.726982	0.723288	0.723988	
FOLD_80_BASE_PENALTY	1.4946	1.56297	1.66478	1.65113	1.61934	1.64793	1.65884	1.64709	1.6828	1.62579	1.66192	1.61456	
PCT_TARGET_BASES_0X	1.0434	1.1106	1.0052	1.0187	1.0174	1.0355	1.0628	1.1328	1.0628	1.1328	0.9995	1.1444	
PCT_TARGET_BASES_1X	0.989566	0.988894	0.989948	0.989813	0.989826	0.989885	0.989645	0.988716	0.989372	0.988672	0.990005	0.988556	
PCT_TARGET_BASES_2X	0.988805	0.988046	0.989149	0.988985	0.989117	0.989044	0.988985	0.987876	0.9885	0.987821	0.989181	0.987729	
PCT_TARGET_BASES_10X	0.985712	0.984704	0.986406	0.985414	0.986356	0.985914	0.985394	0.983991	0.984891	0.984018	0.985777	0.984427	
PCT_TARGET_BASES_20X	0.981338	0.978985	0.983155	0.981637	0.983107	0.981637	0.977926	0.977003	0.977926	0.977003	0.980075	0.979203	
PCT_TARGET_BASES_30X	0.971815	0.965903	0.977446	0.962086	0.977369	0.972638	0.964318	0.963446	0.96801	0.958558	0.966921	0.967118	
PCT_TARGET_BASES_40X	0.951497	0.935317	0.966173	0.926071	0.963751	0.95373	0.924094	0.924039	0.924039	0.915139	0.938671	0.938671	
PCT_TARGET_BASES_50X	0.914024	0.877483	0.945884	0.863798	0.944825	0.919497	0.876613	0.876117	0.861177	0.839284	0.888768	0.885171	
PCT_TARGET_BASES_100X	0.48104	0.4156	0.671693	0.427207	0.657375	0.566831	0.455685	0.455794	0.438207	0.361949	0.481417	0.45344	
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_10X	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_20X	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_30X	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_40X	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_50X	0	0	0	0	0	0	0	0	0	0	0	0	
HS_PENALTY_100X	0	0	0	0	0	0	0	0	0	0	0	0	
AT_DROPOUT	0.0313876	0.0253785	0.0256347	0.0250898	0.0255832	0.0261088	0.0250747	0.0241472	0.0238407	0.0243836	0.0236229	0.0243775	
GC_DROPOUT	8.68775	12.6151	13.7783	13.7143	13.0223	13.4476	13.7376	14.0815	14.5251	13.5466	14.0563	13.5791	